GROUP WORK PROJECT #FE GWP 1 M3 MScFE 610: FINANCIAL DATA

GROUP NUMBER: 7681

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Multicollinearity

Definition

Multicollinearity is present when two or more independent variables in a regression model are correlated. Multicollinearity can lead to skewed or misleading results.

Description

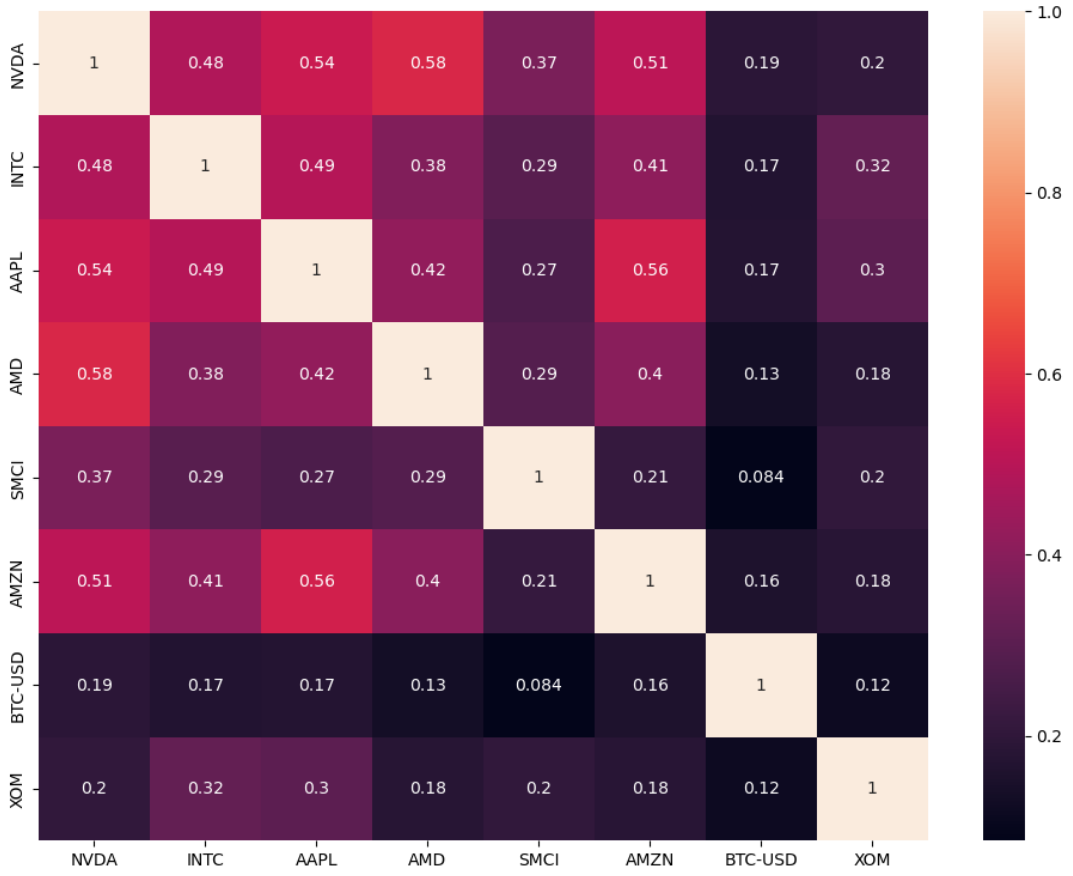
To filter out multicollinearity, we can use correlation matrix and measure the correlation between the variables.

Demonstration

From the adjusted closing prices of Nvidia stock and other 7 assets, we calculated daily returns and created correlation matrix with Pearson coefficient. Additionally, we made Principal Component Analysis on the variables and ran the OLS regression model to test our results.

Diagram

For visualization we used the seaborn heatmap.



Diagnosis

From the correlation matrix, we can identify the presence of multicollinearity as we have more than 2 daily return pairs \ that are correlated (where |Pearson coefficient| > 0.5)

If one seeks independent variables for dependent variable NVDA, we can derive the following conclusion:

- AAPL, AMZN show a strong positive correlation with NVDA, hence these variables might be a good choice to retain for the regression models.
- AMZN and AAPL also show a strong positive correlation with each other, this is a potential warning for multicollinearity.
- The rest of the variables show a moderate/weak correlation with both NVDA and each other, this indicates one should reconsider adding these to the regression models.

Based on Principal Component Analysis we can identify 6 variables from the loadings that cover ~94% of the variance: AAPL, BTC-USD, XOM, SMCI, AMD, INTC After running the OLS regression models with and without AMZN we can get the conclusion that our model did not improve with removing AMZN based on the adjusted R^2 has decreased. This justifies the results of our correlation test that AMZN can be significant in explaining NVDIA returns. VIF index has increased from 1.972 to 2.049 which means there is more multicollinearity in our model if we exclude AMZN.

Damage

The presence of multicollinearity increases the variances, covariances and standard errors of the coefficients in the regression model which lead to wider confidence intervals and results in less reliable statistical inferences. P-values become less reliable in our analysis, and it makes it more complicated to identify each variable's individual effect. The coefficients become very sensitive to small changes in the model.

Directions

- Drop variables based on |Pearson coefficient| > 0.8.
- Run Principal Component Analysis
- Run regression models and consider choosing variable based on VIF formula.

$$VIF_i = \frac{1}{1 - R_i^2}$$

Skewness

Definition

'skewness' is taken to see the 'extent of asymmetry (egyankosh #)' in the data. When the distribution is asymmetrical, it is considered to be "skewed" (egyankosh #) '. A symmetrical distribution has zero skewness.

$$\operatorname{skew}(X) = \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^3\right]$$

Description

When data is skewed there are two possibilities:

1) it is positive skewness or

2) it is negative. When data is bell shaped or unimodal, the direction of the skewness is driven by Mode .The spread of the data on the side of the mode helps in deciding which way is the skewness going to be.

The distribution has skewness if it has :

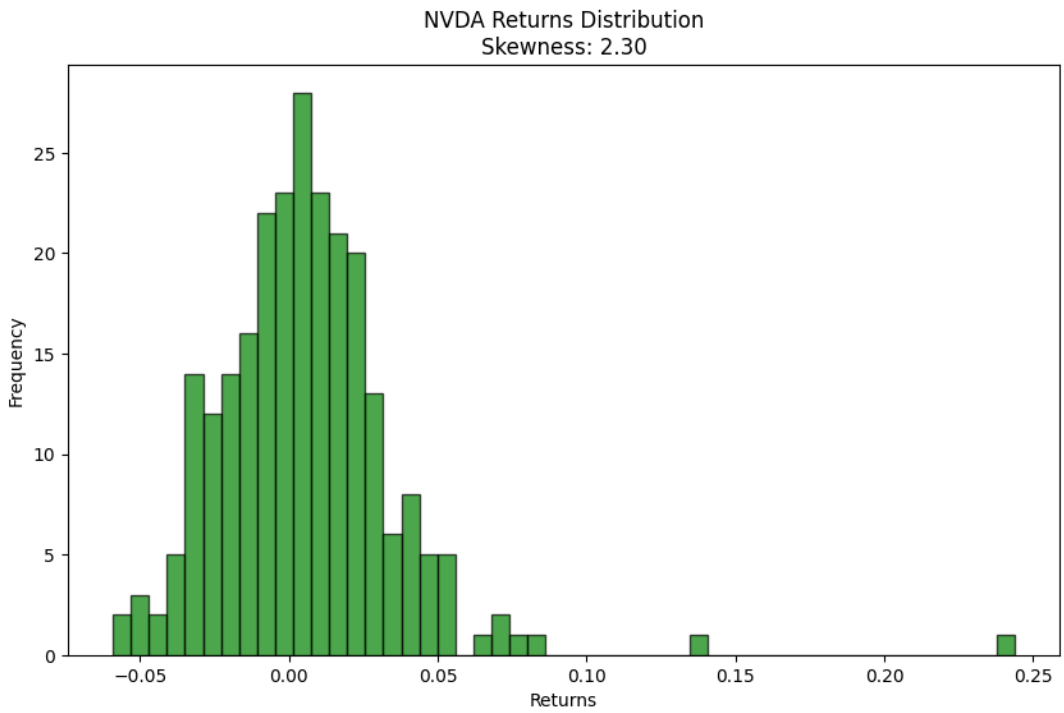
1)there should be no coincidence of mean median and mode.

2) The sum of the positive deviations from the median is not equal to the sum of the negative deviations. The sum taken from the median towards the positive side should not be equal to the sum taken from the negative side from the medium

3) The spread of the frequency and the spread on either side of should not be equal

4) Taking the distance from the median the quartile should not have equal distance

Diagram



Diagnosis

To analyse the extent of asymmetry , some measure of skewness are taken ,one is of absolute and the other id relative type . Absolute Measures – it tells how much the asymmetry is present to compare more than 2 distributions, coefficient of skewness is taken . to measure skewness which is relative : median and the interquartile range,are taken . 1) Karl Pearson's Coefficient of Skewness. This is calculated for the first absolute measure

$$Sk_1 = \frac{\bar{X} - Mo}{s}$$

Where $\bar{X}$ is mean , Mo = the mode and s = the standard deviation for the sample in concern

Some can also use non-parametric tests, such as the Mann-Whitney U test or the Kruskal-Wallis Test which has assumption of normal distribution.

Damage

Misleading Mean Return: In a skewed dataset, the mean return is disproportionately affected by the few extreme positive returns. For example, if most daily returns are between -2% and 1%, but there are a few outliers at +10%, the mean may suggest the stock performs better than it typically does.

Risk Misrepresentation: Skewness can understate the downside risk of a stock. For instance, if risk metrics like Value at Risk (VaR) (Grigoletto, 2006) assume a symmetric distribution, they might underestimate the likelihood of negative returns and overestimate the probability of positive returns.

Modeling Errors: Financial models like the Capital Asset Pricing Model (CAPM) assume normally distributed returns. Significant skewness can violate this assumption, leading inaccurate estimates of beta and expected returns, affecting portfolio optimization decisions

Directions

Transformations help to make skewed data more symmetric, improving the reliability of statistical analyses and meeting the assumptions of many models like GARCH, ARIMA ETC (Grigoletto, 2006)

We have used log transformations and as well as box cox transformations and the data had become somewhat normally distributed leading for us to apply.

Skewness of Raw Data: 2.30498

Skewness of Log-Transformed Data: 2.06440

Skewness of Box-Cox Transformed Data: -0.10037

Sensitivity to outliers

Definition

The sensitivity to outliers indicates that a statistical or machine learning model would not be robust to a dataset's extreme values (outliers). By definition, an outlier is a point that is much different from the other observations, and its detection is based on math with the help of Z-scores, Interquartile Range (IQR)

Z-Score Formula:

$$Z = \frac{x - \mu}{\sigma}$$

Where, x is the data point, μ is the mean, and σ is the standard deviation.

IQR Method:

$$IQR = Q3 - Q1$$

Outliers are identified as data points where:

$$x < Q1 - 1.5 \times \text{IQR} \quad \text{or} \quad x > Q3 + 1.5 \times \text{IQR}$$

Description

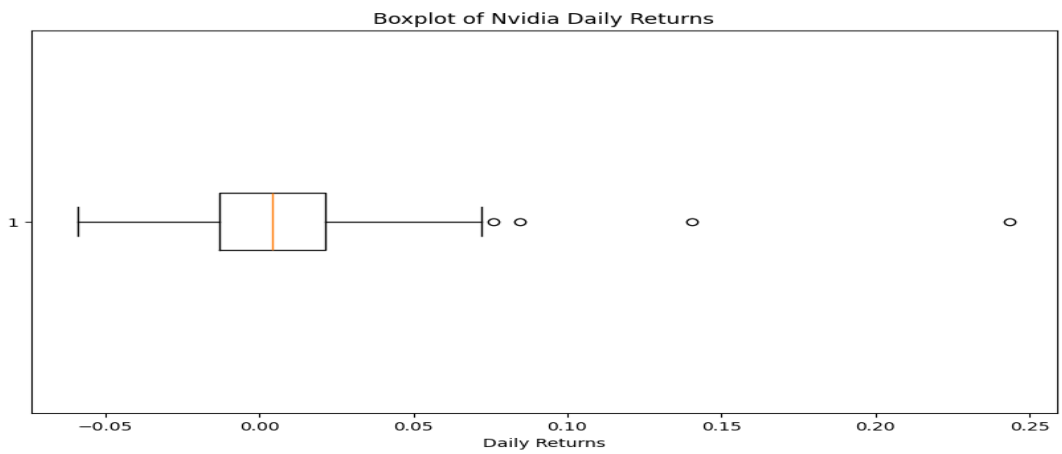
Outliers are extreme values in the data that differ significantly from other observations. They can exacerbate our mean and standard deviation, which can make our models less robust.

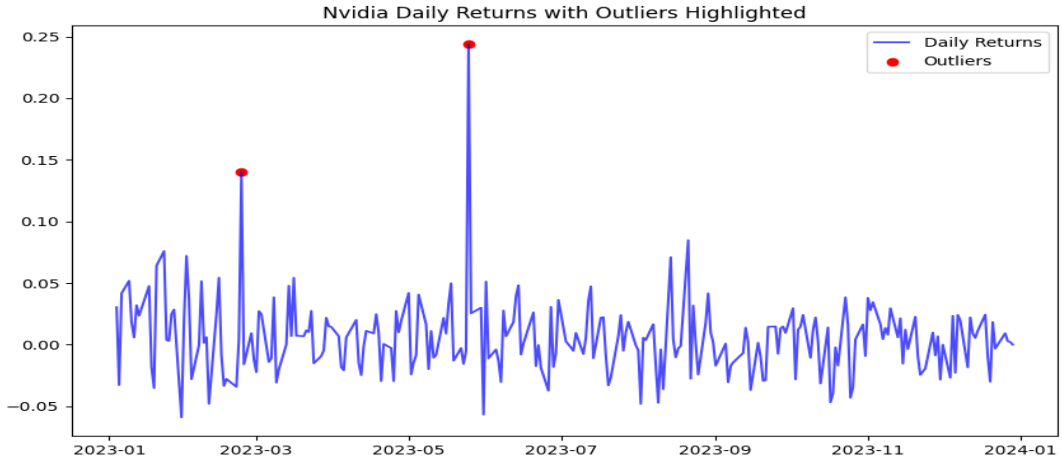
Demonstration

From the adjusted closing prices of Nvidia stock, we calculated daily returns. We filtered the outliers as well with Z-score and IQR and plotted in time-series. (Please find in the Python file)

Diagram

Creating the boxplot of the distribution of daily returns, including outliers, and highlight these outliers in the time-series plot of daily returns.





Diagnosis

To identify outliers, apply both the Z-score and the IQR methods. After identifying these outliers, assess their impact by comparing summary statistics, such as the mean and standard deviation, before and after addressing the outliers.

Damage

There is a wide range of ways that outliers can affect the analysis as well as decision-making processes.

They can result in misleading and unrepresentative statistical summary numbers, like the mean or variance. Such extreme values are then misrepresented, which can lead to poor performance of the model, where algorithms cannot fit these extreme values well or predict skewed results.

Furthermore, in finance data sets, including outliers, may lead to wrong decisions by causing wrong estimates of volatility or prices and defrauding investors or stakeholders. Handling outliers is crucial for accurate analysis, and that ensures quality decision-making.

Directions

The team implemented a number of techniques to mitigate the problem of outliers, which were used to analyze Nvidia's stock returns data and observe the impact of each technique on time series modelling. We identified outliers using two methods: Z-score ($|Z| > 3$) and IQR (values outside $1.5 \times$ IQR from the first and third quartiles). Visualization through a boxplot revealed extreme daily return values, confirming their presence. Winsorization, a robust technique to cap extreme values at the 5th and 95th percentiles, was applied to handle outliers. We performed a linear regression model on the time variable as the predictor and derived the R-squared and RMSE metrics to test the impact of winsorization. The results indicated that by winsorizing, model performance was improved by lower volatility (standard deviation) and higher predictive accuracy (lower RMSE and higher R-squared). It shows that winsorization is a successful technique to compensate for the negative effect of outliers without distorting the overall dataset. (Hadi et al, 1993)

Challenge: Integrated Approach to Time Series Modeling Challenges

The section is of time series modeling challenges, highlighting the interconnected nature of statistical complexities in financial forecasting.

Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model, models the conditional variance and it is represented where $yt = \mu + \epsilon t$ and ϵ has conditional variance in it $\sigma_t z_t$.

μ_i Mean return
ϵ_t: Residual
σ_t: Conditional standard deviation (volatility) at time ttt
z_t: A white noise error term with zero mean and unit variance

Conditional variance equation:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2$$

σt^2: Conditional variance (forecasted volatility) at time t
α_0: Constant term
α_i: Parameters for lagged squared residuals (ARCH terms)
β_j: same as above but parameter for lagged conditional variances (GARCH terms)
p: number of lags for garch terms
q: number of lags for arch terms

Skewness:

Description: Returns show positive skewness (0.676) affecting model assumptions
Demonstration: NVDA returns skewness = 0.676, indicating right-tailed distribution
Diagnosis: Implemented through skewness detection and Box-Cox transformations
Damage: Biased volatility estimates and risk metrics

Outlier Sensitivity:

Description: GARCH parameters show high sensitivity to extreme values
Demonstration: Beta[1] = 0.97 creating strong persistence
Diagnosis: Implemented IQR and z-score based detection
Damage: Volatility overestimation during extreme events

Overfitting:

Description: Complex model structure vs. available data points
Demonstration: Model shows high R-squared despite simple structure
Diagnosis: Cross-validation and information criteria (AIC: -2815.16)

Damage: Poor out-of-sample forecasting

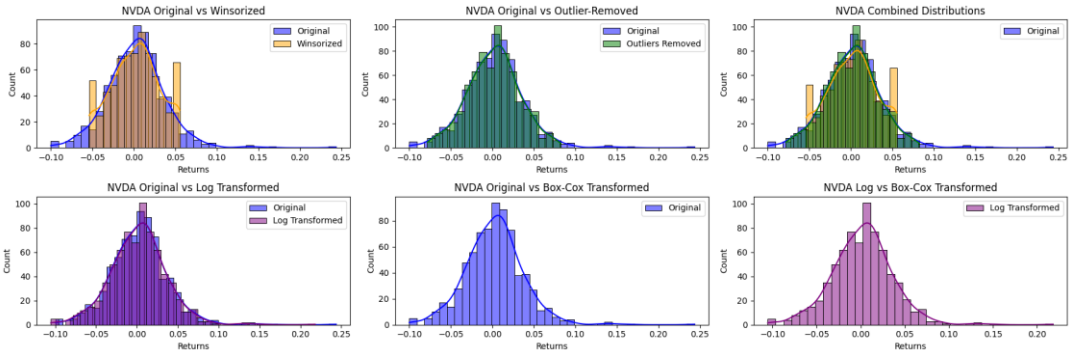
Interpretation:

Description: Complex parameter relationships in GARCH models
Demonstration: Multi-level transformations and preprocessing steps
Diagnosis: Visualization comparisons across different transformations
Damage: Reduced model interpretability

Factor Prioritization:

Description: Multiple competing variables affecting volatility
Demonstration: Various preprocessing methods
Diagnosis: Comparative analysis of different transformations
Damage: Suboptimal feature selection

Visualisation



GARCH Model Statistical Challenges

Skewness WARNING

Right-tailed distribution affecting model assumptions

Severity: 67.6%

Impact: 80.0%

Parameter Persistence CRITICAL

High Beta[1] indicating strong volatility persistence

Severity: 97.0%

Impact: 90.0%

Model Fit WARNING

AIC: -2815.16, indicating potential complexity concerns

Severity: 75.0%

Impact: 70.0%

Solutions to Key Challenges

Outlier Sensitivity

- Use robust detection methods:
- IQR: Flag and adjust data beyond 1.5 times the interquartile range.
- Z-Score: Identify observations where |z| > 3.
- Data quality is done through visualizing comparison of the raw and the adjusted datasets.
- AS mentioned in one of the challenges in the project
- Applying winsorization to smooth extreme values, gives better results (Nanchang University, China et al.)

Managing Overfitting

- Model structures are simplified by reducing garch lag parameters(p,q)
- Simple model structures by reducing GARCH lag parameters (p, q).
- Use rolling or expanding windows for cross-validation.
- Evaluate models with out-of-sample metrics like MAE or MAPE.
- Select models based on AIC or BIC to ensure balance between simplicity and fit.

Improving Interpretability

- Consistently apply transformations, such as logarithmic scaling or Box-Cox adjustments.
- Generate side-by-side visual comparisons of pre- and post-transformation data.
- Document all preprocessing steps in detail to maintain transparency.

Prioritizing Relevant Factors

- Assess feature importance using techniques like SHAP values or permutation importance.
- Test and compare various preprocessing strategies, including scaling and outlier handling.
- Use sequential feature selection to identify the most predictive variable subsets.
- Addressing Stationarity and Variance Issues
- Conduct ADF tests to confirm stationarity, applying differencing if necessary.
- Stabilize variance through Box-Cox transformations.
- Regularly monitor rolling variance to detect heteroscedastic patterns.

Enhancing Model Robustness

- Rescale input data to address parameter convergence warnings (e.g., multiply the series by 10).
- Explore alternative modeling approaches, such as EGARCH, TGARCH, or FIGARCH.
- Validate models with both synthetic and real-world datasets to test reliability.

REFERENCES

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